SEOP

Seed to Ontology Pipeline (SEOP)

SEOP Enterprise Ontology Intelligence Platform

CMMI Level 5 · BURR-Aligned · 46 Modules · 7 Pipeline Layers · 7 Cross-Cutting Service Groups

CMMI L5 | 46 Modules | 7 Pipeline Layers | BURR-Aligned | Dual Metrics | 4 Prompt Tiers | Hybrid LLM+Rule | R2RML/D2RQ/RML | 10k Bootstrap CI | Nemenyi/CD Diagrams | 14+ Standards | 100% Provenance | RBAC/ISO 27001 | Data Governance | Observability | Enterprise Analytics | Managed Lifecycle

Standards Mapping Catalog (M10)

Generic Vocabularies
- Schema.org · DCAT · DCTERMS · SKOS · PROV-O

Geospatial & IoT
- OGC GeoSPARQL · WGS84 · W3C SSN · SOSA · SAREF

Financial & Domain
- FIBO · SNOMED CT · UMLS

Top-Level Ontologies (ISO-Aligned)
- BFO · DOLCE · ISO/IEC 21838 · ISO 29002

Relational Mapping Standards
- R2RML (preferred for DB Ontology)
- D2RQ (legacy) · RML (heterogeneous)
- Direct Mapping (W3C) — baseline

Domain Bridges
- Financial ↱ FIBO · Geospatial ↱ GeoSPARQL
- IoT ↱ SOSA/SSN/SAREF · Climate ↱ Multi
- Healthcare ↱ SNOMED/UMLS

Consumption
- M06 Upper Onto · M09 Conformance · M44 Mapping
- M16 Data Quality · M39 BURR Benchmark
- Drift detection · coverage reporting per release
- Versioned alignment files (SemVer) · reusable

SEOP Usage
- Every concept mapped to ≥ 1 standard
- Standard drift reported to M19 Governance
- Coverage % is a hard release gate
- Feeds M40 standards-conformance metric

Version History
- SemVer-managed in M28 Registry
- Change approvals via M05 DAR · M36 CM

Data Foundation (M15–M20)

M15 Data Ingestion Framework
PDF · JSON · XML · CSV · SQL · KG · Multimodal parsers
Real-time + batch · Schema introspection · Provenance per chunk
Connector marketplace · Retry · Feeds M42 routing signals
Emits FK presence, normalization degree, schema entropy ↱ M42

M16 Data Quality Framework
6 dimensions: completeness · accuracy · consistency · timeliness
validity · uniqueness · Rule engine · Anomaly detection
DQ scorecards · Threshold quarantine · Pre-Layer-2 gate
BURR insight: extra sampling on denormalized tables

M17 Data Dictionary Service
Business glossary · Technical schema · Synonyms · Lineage
Stewardship · Approval workflow · PII tags
Canonical naming normalizer (`_` ↱ `-`) for M40
Feeds M05 disambiguation · M40 lexical scoring

M18 Data Lifecycle Management
Retention · Archival · Purge (audited) · Tiered storage
Legal hold · Right-to-be-forgotten (GDPR/CCPA)
Data versioning · Lineage preservation across deletions
Preserves BURR scenario artifacts (M45) for regression

M19 Data Governance Center
Ownership · Stewardship · Policies · Standards
Classification (PII, confidential) · Access reviews
Policy-as-code · Compliance dashboards · Lineage governance
Approved terms ↱ M17 · Audit ↱ M27

M20 Metadata & Ontology Service
Central metadata catalog (technical + business + operational)
Ontology artifact registry: TBox · ABox · SHACL · R2RML mappings
Search · REST + GraphQL + SPARQL APIs
Feeds M39 BURR benchmark with reference artifacts

BURR-Informed Data Signals

Signals Emitted to M42 (Model Router)
↱ DB normalization degree (1NF–BCNF)
↱ Foreign-key presence / absence
↱ Table cardinality (N:M · 1:N · hierarchies)
↱ Cross-scenario composition flag
↱ Denormalization index ↱ M43 trigger
↱ Naming-scheme drift ↱ M17 normalizer
↱ FK-derived relation confidence boost
↱ DQ score by dimension ↱ M16 gate
↱ Enterprise-scale detection (schema entropy)
↱ Prompt-length proxy (schema token count)
↱ All signals carry PROV-O provenance (M27)

CMMI-5 Hooks
- CAR (M24) · signal anomalies ↱ root-cause
- MA (M30) · signal ingestion ↱ measurement
- QPM (M33) · control charts per signal

SEOP Value Narrative

Trust
Every element traced from seed ↱ source ↱ axiom ↱ simulation
PROV-O decision chains · dual-metric quality · human sign-off

Grounding
Statistical evidence per candidate (M04) · DQ gates (M16)
BURR scenario categories drive coverage guarantees

Standards Alignment
14+ vocabularies · ISO/IEC 21838 · ISO 29002
R2RML-first for DB Ontology mappings

Continuous Evolution
Simulation errors ↱ patch ↱ human review ↱ regression ↱ deploy
Statistical significance required before promotion

Scientific Rigor
Dual metrics mandatory · 10k bootstrap CI · Nemenyi
Critical-difference diagrams · evidence-grade decisions

CMMI-5 Ready
22 PAs mapped to modules · SCAMPI-appraisal ready
Auto-collected evidence across all 46 modules

Enterprise-Ready
RBAC · PII redaction · SOC 2 · ISO 27001 · GDPR · HIPAA

L0 · Seed Formalization & Competency-Question Studio (M01)

Seed Contract
Competency Questions · Scope · Vocabulary constraints · Acceptance criteria
Intended reasoning tasks · Target simulation · Upper-ontology commitment (BFO/DOLCE)
Human approval required before pipeline execution

Versioned Seed Registry
Immutable snapshots · Change log · Approval workflow · Project/domain/regulatory context
Feeds Evaluation Harness (M29) and BURR Benchmark (M39)
Referenced by all downstream layers via immutable seed ID

L1 · Data Ingestion Framework & Contextual Analysis (M15 · M02)

Multimodal Ingestion (M15)
PDFs · Reports · JSON · XML · CSV · SQL · KG
Parsers: LlamaParse, Unstructured.io · OCR · Tables
Real-time + batch · Retry & backoff · Provenance per chunk
Emits FK/normalization/entropy signals to M42
Rejected chunks ↱ DQ quarantine (M16)
DB normalization degree computed for M42
Publishes lineage edges to M19 Governance

Simulation Requirement Analyzer (M02)
Boundaries · Agents · Variables · Constraints · KPIs
Digital-twin vs. ABM vs. hybrid · Time horizon
Elicits implicit requirements via LLM
Drafts CQs ↱ Layer 0 for human review
Links simulation intent to ontology scope
BURR insight: composite scenarios need explicit CQ coverage
Feeds M45 Scenario Library when replayable

Metadata & Schema Profiling
Schema introspection · API payload sampling
Table/column profiling · Range & distribution stats
Feeds M16 DQ · M20 Metadata · M17 Dictionary
Detects N:M / 1:N / hierarchy patterns ↱ M42
CQ elicitation drafts ↱ Layer 0
Publishes schema characteristics to M42 routing
Emits token-count proxy ↱ M31 prompt-length guard

L2 · Real-World Seed Extraction (M03 · M04 · M05)

Entity & Relation Mining (M03)
NER · RE · GLiNER · LLM prompts
Candidate classes, properties, instances
Provenance + confidence per candidate
Evidence spans preserved · Human spot-check (M12)
Uses M17 Dictionary canonical terms
DQ-scored inputs only (M16 gate)
BURR: FK-derived pairs boost relation confidence
Naming canonicalized before F1 scoring

Statistical Grounding (M04)
Cross-reference with DB columns · API payloads
Schemas · Data ranges · Coverage metric per seed
Rejects hallucinated elements (BURR finding)
Evidence count + diversity score
Uses M16 DQ profiles · M20 metadata
Rejected ↱ M27 Provenance + DQ feedback
Mapping-coverage target set per element
Feeds M46 mapping-based metric

Disambiguation & Clustering (M05)
Embedding similarity · HDBSCAN · FAISS/Qdrant
Canonical term resolution against M17
Synonym merge proposals · Human-approved (M12)
Naming normalizer (`_` vs `-`) handled pre-metric
Versioned seed output
Drift alerts ↱ M21 Data Observability
Approved merges committed to M20
Canonical form feeds M40 lexical scoring

L3 · LLM-Driven Ontology Synthesis (M06 · M07 · M08 · M41 · M44)

Upper Ontology Alignment (M06)
Fixed: BFO or DOLCE per project
Explicit mapping rules
Endurant vs. Perdurant checks
Violation detection · Review (M12)
Human sign-off (high-impact)
Uses M19 Standards Catalog
D2RQ/RML syntax checks
CMMI-5 PPQA

Domain Ontology Synthesizer (M07)
Subclasses · Object & data properties
Restrictions · Axioms · Disjointness
Decision chain per axiom
Structured diff ↱ human review (M12)
Uses M17 canonical terms
Injects M41 prompt tier per task
Calls M42-selected model (LLM or rule)
Emits TBox + candidate mappings
Committed to M28 after validation
Feeds M43 hybrid if denormalized

Serialization (M08) + Mapping (M44)
Turtle · RDF/XML · OWL Manchester
R2RML (default for DB Ontology)
D2RQ (legacy) · RML (heterogeneous)
Direct Mapping (W3C baseline)
SHACL shapes (draft) · SWLs (draft)
Namespace & IRI conventions
BURR: R2RML preferred over RML
Committed to M28 registry
Feeds M39 BURR benchmark runner
Feeds M20 metadata search

Prompt Strategy Engine (M41)
Tier 1 · Simple (baseline)
Tier 2 · Enhanced (example anchor)
Tier 3 · Fewshot (3 scenarios)
Tier 4 · Checklist (fewshot + validation)
Versioned prompt artifacts
Auto-tier selection per task complexity
BURR: Tier 4 best statistically
Open-models force Tier 3+ (Qwen fix)
Prompt-length warning ↱ M31
Feeds M40 prompt-quality attribution

L4 · Standardization & Multi-Tier Validation (M09 · M10 · M12)

Syntactic (M09)
RDF/XML / Turtle well-formed
Namespace checks
RDFLib / Jena parse
Errors ↱ M27
R2RML parser validation
D2RQ/RML syntax checks
CMMI-5 PPQA

Structural (M09)
SHACL / SWLs validation
Cardinality · Datatype
Class disjointness
Domain / range checks
Mapping coverage check
BURR metric integration
CMMI-5 PPQA · VER

Logical (M09)
HermiT / Pellet / ELK
Owlready2 consistency
Unsatisfiable classes
Disjointness violations
Explains / justification
Mapping consistency
CMMI-5 PPQA · VAL

Quality (M09)
OOPS! / OntoClean
Pitfall detection
Naming conventions
Rigidity / identity
Annotation completeness
Lexical F1 (M40)
CMMI-5 PPQA

Standards Conformance (M10)
Schema.org · DCAT · DCTERMS · SKOS
PROV-O · GeoSPARQL · WGS84 · SSN/SOSA
FIBO · SAREF · BFO/DOLCE
R2RML / D2RQ / RML conformance
ISO/IEC 21838 · ISO 29002
Coverage % · drift detection
CMMI-5 DAR · VER

Human Review Gate (M12)
Triage by severity · Approve / reject / edit · Dual review for schema changes · Decision ↱ M27 · Gate before ABox (M11) and Simulation (M13)
Escalation SLAs · SME sign-off · Feeds M22 Success Metrics · M23 KPI Framework · M40 Dual-Metric scoring

L4.5 · ABox Population & SPARQL CQ Testing (M11)

Instance Extraction & Linking
ABox entities linked to TBox classes
Entity resolution across sources
Provenance per instance · Link prediction (LLM)
Confidence scoring · Review (M12) · CMMI-5 PPQA · VER

SPARQL CQ Tests
Every CQ from Layer 0 executed
Pass / fail / partial · Templates per CQ
Results ↱ M29 Evaluation Harness
BURR metric computation triggered · CMMI-5 VER · VAL · MA

Coverage & Gap Report
Per-CQ coverage %
Missing classes / properties
Suggested patch seeds ↱ Layer 5
Trendline across versions · Published to M24 · CMMI-5 MA · QPM

L5 · Simulation Feedback, Patch Governance & Evolution (M13 · M14)

Deployment & Simulation (M13)
Digital twin · Agent-based model
Hybrid simulation · Neo4j/GraphDB
SPARQL endpoint exposed
Version pinned · Rollback ready
RBAC-scoped access via M26
Deployment events ↱ M23 KPI
BURR scenarios replayable here
CMMI-5 CM · PI · CM audit trail per deploy
Agent runtime consumes via SPARQL 1.1

Error Monitoring
Missing-link alerts · Rigid-structure
CQ execution regressions
Simulation KPI drift
Agent decision failures
Feedback ↱ M11 gap report
Threshold escalation ↱ M24 CAR
BURR-style regression detection
CMMI-5 PMC · CAR
Root cause ↱ M24 CAR

LLM Patch Proposals (M14)
Agent analyzes error logs
Proposes new properties / axioms
Constraint relaxation
Impact analysis vs. active sims
Draft in M28 registry
Never auto-applied
BURR-micro: A/B against scenarios
CMMI-5 DAR · CAR · Rationale ↱ M27
M46 significance gate before promotion

Patch Governance (M14)
Proposal ↱ Impact Analysis
↱ Regression Test Suite
↱ Human Approval (M12)
↱ Versioned Release (M28)
↱ Deploy (opt-in) · Rollback
↱ BURR regression gate
CMMI-5 CM · DAR · PPQA
Full audit trail per patch
Reject if BURR score < baseline

Continuous Learning Loop
Sim errors ↱ patches ↱ human review ↱ versioned ↱ re-deploy · regression-tested via BURR (M39) · dual-metric-scored (M40) · KPI-tracked (M23) · CQ-tested (M11)

Enterprise Consumers & Integration

Agentic AI Runtime
SPARQL / GraphQL query
Ontology-driven tool selection
Policy-aware planning
CQ-linked capability discovery
RBAC via M26
Usage ↱ M25 analytics
BURR-scored capabilities only
Throughput ↱ M23 KPI

Digital Twin Platform
Semantic simulation engine
Operational semantics
Missing-link alerts ↱ Layer 5
KPI drift monitoring
Deployment via M13
Feedback ↱ M21 observability
BURR replay on deploy
CM audit trail

BI & Analytics Consumers
Self-service BI (M24)
Executive dashboards (M22)
Operational KPIs (M23)
Adoption (M23)
Observability (M21)
Governance (M19)
BURR scorecards (M40)
Statistical reports (M46)

External APIs
SPARQL 1.1 endpoint
GraphQL API
REST / OpenAPI
Webhooks · Kafka events
Standards-conformant exports
R2RML/D2RQ/RML delivery
RBAC enforced at gateway
BURR score in response header

Analytics, Observability & Measurement (M21–M25)

M22 Data Observability Platform
Pipeline health · Drift · Volume anomalies · Quality signals · Lineage · Alerting · Feeds M23, M24

M22 Success Metrics Framework
Business KPIs (OKRs, goals) · Metrics taxonomy · Attribution to pipeline modules · BURR-quality roll-up

M23 KPI Metrics Framework
SLIs · SLOs · Error budgets · Real-time computation · Historical trending · CMMI-5 MA

M24 Analytics Platform
Self-service BI · Dashboards · Ad-hoc query · Report builder · Embedded analytics · RBAC-scoped

M25 Visitor & Usage Analytics
User behavior · Feature adoption · Funnel · Privacy-first consent-based · Adoption KPIs ↱ M22

Enterprise Foundation Services (M26–M31)

M26 RBAC & Identity Access Management
OAuth2/OIDC · Roles, groups, permissions · Row/column/attribute-level access
Granular ACLs on ontology artifacts, data sources, dashboards · JIT access · MFA
Governs access to BURR scenarios (M39) and statistical reports (M46)

M27 Provenance & Audit Service
PROV-O-aligned decision chain for every class, property, axiom, mapping
Records source, prompt tier (M41), model (M42), reviewer, result
Immutable event log · Query API · Compliance evidence packages

M28 Ontology Registry & Versioning
SemVer · Diff · Rollback · Dependency graph · Immutable releases
Git LFS / DVC · Metadata publication to M20
BURR scores archived per version · Deployment gate for M12

M29 Evaluation Harness
Composite KPI dashboard · 4 families: ontology quality, standards, simulation, human
Integrates M40 dual-metric results · M46 statistical reports
CMMI-5 MA + QPM integration · Trend across releases

M30 Security & Compliance Center
PII redaction (Presidio) · Vault · Encryption at rest/transit
SOC 2 · ISO 27001 · GDPR · HIPAA · Audit logging
Incident response · Posture dashboards · CMMI-5 support

M31 LLM Agent Orchestrator
Prompt versioning · Retries · Guardrails · Cost · LangGraph/Prefect
Multi-agent · Model routing (uses M42) · Prompt drift detection
Open-model fewshot injection (BURR fix) · Failover · LangSmith integration

Model Routing & Hybrid Pipeline (M42 · M43)

M42 Model Routing & Selection Service
Per-task routing driven by BURR signals: DB normalization, FK presence, N:M density, schema entropy
LLM (GeminiGPT/Qwen) vs rule-based (RDB2 Onto/Ontop) selection · Confidence threshold
Fallback chain · Per-model cost/latency trade-off · Prompt-length guard · Decision ↱ M27

M43 Hybrid Ensemble Pipeline
Runs rule-based + LLM in parallel; fuses results per-element using BURR-informed confidence
Selects best per-element (CRA) · Explains fusion via M27
Triggered when M42 flags denormalized or composite scenarios (BURR "Special")

CMMI Level 5 Process Areas (M32–M38)

M32 Organizational Process Performance (OPP)
Baselines · Capability baselines · PPMs · Targets per project · Consumes M38, M22

M32 Quantitative Project Management (QPM)
SPC control charts · Objectives management · Sub-process analysis · Predictability

M34 Causal Analysis & Resolution (CAR)
Root cause on defects/regressions/incidents · 5 Whys/Fishbone/Fault tree · Preventive changes

M35 Decision Analysis & Resolution (DAR)
Formal alternatives evaluation (patch, standards, LLM choice) · Criteria weighting · Rationale ↱ M27

M36 Configuration Management (CM)
Baseline ID · Control · Status accounting · Audits · Applies to all pipeline artifacts

M37 Process & Product Quality Assurance (PPQA)
Objective evaluation against standards · Non-compliance tracking · Escalation

M38 Measurement & Analysis (MA)
Measurement objectives · Metric spec & collection · Analysis · Foundational for OPP, QPM, CAR

CMMI-5 Continuous Representation — enforced by M27+M29+M38
All 22 PAs mapped to modules · Evidence auto-collected · SCAMPI-appraisal ready
BURR metrics become primary MA/OPP/QPM data stream — auditable at every release

SEOP Brand Commitments

Trust
Every ontological element traceable from seed ↱ source ↱ axiom ↱ simulation ↱ consumer.

Grounding
Statistical evidence per candidate (M04) · DQ gates (M16) · BURR scenario categories.

Standards
14+ vocabularies · ISO/IEC 21838 · ISO 29002 · R2RML-first DB Ontology mappings.

Rigor
Dual metrics mandatory · 10k bootstrap CI · Nemenyi · critical-difference diagrams.

Evolution
Simulation errors ↱ patches ↱ human review ↱ regression ↱ deploy. Never auto-applied.

Enterprise-Ready
RBAC · PII redaction · SOC 2 · ISO 27001 · GDPR · HIPAA · CM audit trail.

CMMI-5
22 PAs mapped to modules · evidence auto-collected · SCAMPI-appraisal ready.

Scientific
Every quality claim passes M46 significance before promotion in M14 / M31 / M42.

SEOP BURR Evaluation Suite (M39 · M40 · M41 · M44 · M45 · M46) — the measurement & regression engine

BURR Learnings Applied in SEOP
Dual metrics mandatory (Lexical + Mapping) · 4 prompt tiers versioned · LLM vs rule routing · Open-model fewshot injection · R2RML preferred · Naming normalization · Bootstrap CI + CD diagrams · Scenario categories (Basic / Attributes / Normalization / Relationships / Hierarchies / Special) · Enterprise-scale and composite flagged

M39 BURR Benchmark Harness
54 micro-benchmark scenarios (6 categories) · 4 real-world scenarios (Mondial, Mondial-FK, iSWC, NPD, Enterprise)
Runs ontology learning systems against gold-standard mappings
Configurable datasets: BURR-provided, customer-specific, scenario library (M45)
Triggered on every release; regression gate in M14 patch governance
CMMI-5 VER · VAL · MA · QPM — evidence captured automatically
Exports per-system × per-scenario × per-element score matrices

M40 Dual-Metric Evaluation Engine
Two mandatory metric families per element (CRA/A): Lexical F1 + Mapping-based F1
Auto naming normalization (`_` ↱ `-`) before scoring
Detects hallucination signals — high lexical, low mapping (BURR enterprise scenario)
Reverse detection — low lexical, high mapping (minor naming drift)
Per-model, per-prompt-tier, per-category scorecards
Feeds M22 Success Metrics, M29 Evaluation, M23 QPM

M41 Prompt Strategy Engine
4 tiers: Simple · Enhanced · Fewshot · Checklist (BURR's winning tier)
Versioned prompt artifacts · A/B testing framework
Auto-tier per task complexity · LLM-assisted prompt refinement
Open-model override: force Fewshot+ for Qwen/Mistral (0.07/0.38 fix)
Checklist validation enforced before response acceptance
Per-tier quality reported to M40 · Attribution in M27

M44 Mapping Format Generator
R2RML — default for DB Ontology (BURR recommendation)
D2RQ — legacy interoperability support
RML — heterogeneous source integration
Direct Mapping (W3C) — baseline for validation
Round-trip validation: parse + query + verify
Feeds M39 benchmark · Delivered via external API

M45 Scenario Library & Challenge Bank
Versioned repository of micro-scenarios (Basic, Attributes, Normalization, Relationships, Hierarchies, Special) and real-world scenarios
Includes BURR's 54 micro + 4 real-world; extended with enterprise-scale and composite scenarios
Customer-uploadable scenarios · Gold-standard mapping preservation · Governance via M19
Feeds M39 benchmark runner and M13 simulation deployment · RBAC-scoped via M26
Scenario growth is a KPI (M23) · Scenario coverage reported in M24 dashboards
CMMI-5 CM baseline · Growth managed by DAR (M18)

M46 Statistical Rigor Service
10,000-sample bootstrap for 95% confidence intervals (BURR exact methodology) · Nemenyi post-hoc test · Friedman test
Critical-difference diagrams (Demšar) · Significance-adjusted ranking · Automated report generation
Per-element CIs (Concept / Relationship / Attribute) · Per-scenario comparison · Cross-version delta analysis
Statistically-significant "winner" detection before promoting any patch/model/prompt in M14/M31/M42
Publishes to M24 dashboards · Feeds M32 QPP baselines · Reviewed under M38 MA · Evidence ↱ M27
CMMI-5 MA, QPM, OPP — statistical rigor is a CMMI-5 prerequisite; M46 makes it automatic